SCHOOLBRIDGE

WhatsApp-First School Operations & Revenue Automation Platform

USER STORIES — DEVELOPMENT BACKLOG

Ready for Sprint Planning & Development

Version 1.0 | 2025

MUST HAVE — MVP	SHOULD HAVE — v1.1	NICE TO HAVE — v2.0

SchoolBridge — Development User Stories

This document contains all user stories for the SchoolBridge platform, organised by epic. Each story follows the standard format: As a [role], I want to [goal], so that [benefit]. Every story includes acceptance criteria, priority (MoSCoW), and story point estimate for sprint planning.

Stories marked MUST HAVE form the MVP backlog. SHOULD HAVE stories target v1.1. NICE TO HAVE stories are planned for v2.0 and beyond.

Backlog Summary

Epic	Title	Stories	MoSCoW Split
EPIC-01	Authentication & User Management	**7**	5 Must / 2 Should
EPIC-02	School & Student Administration	**8**	5 Must / 3 Should
EPIC-03	Fee Collection & Revenue Automation	**10**	7 Must / 3 Should
EPIC-04	Mobile Money Integration	**6**	6 Must
EPIC-05	WhatsApp Communication Layer	**9**	6 Must / 3 Should
EPIC-06	Academic Module & Report Cards	**8**	5 Must / 2 Should / 1 Nice
EPIC-07	Attendance Management	**5**	3 Must / 2 Should
EPIC-08	Director & Analytics Dashboard	**6**	2 Must / 3 Should / 1 Nice
EPIC-09	Automation Engine	**5**	4 Must / 1 Should
EPIC-10	Offline & Connectivity Resilience	**4**	2 Must / 2 Should
EPIC-11	Security & Compliance	**5**	5 Must
EPIC-12	Notifications & Alerts	**4**	2 Must / 2 Should

Total stories: 77 | Must Have: 47 | Should Have: 24 | Nice to Have: 6

EPIC-01 Authentication & User Management

US-001 — Create a new school account with name, location, and contact details

Story ID	US-001
As a...	System Admin
I want to...	create a new school account with name, location, and contact details
So that...	the school is onboarded as an isolated tenant with its own data space
Priority / Points	**Must Have** | Story Points: 3
Acceptance Criteria	School created with unique ID; no cross-tenant data visible; onboarding confirmation email sent to director; duplicate school name flagged
Dev Notes	Multi-tenant row-level security in PostgreSQL; school_id scopes all queries

US-002 — Log in with my email and password and receive a JWT access token

Story ID	US-002
As a...	School Director
I want to...	log in with my email and password and receive a JWT access token
So that...	I can securely access the admin portal and my session is time-limited
Priority / Points	**Must Have** | Story Points: 2
Acceptance Criteria	Login returns 200 with access token (24hr expiry) and refresh token; incorrect credentials return 401; account locked after 5 failed attempts for 15 minutes
Dev Notes	bcrypt password hashing; refresh token rotation on each use

US-003 — Create staff accounts and assign them a role (Bursar, Teacher, Admin)

Story ID	US-003
As a...	School Director
I want to...	create staff accounts and assign them a role (Bursar, Teacher, Admin)
So that...	each staff member only sees and does what their role allows
Priority / Points	**Must Have** | Story Points: 3
Acceptance Criteria	Roles: Director, Bursar, Teacher, SystemAdmin; Teachers cannot view fee records; Bursars cannot enter marks; Director sees everything; role change logged in audit trail
Dev Notes	Django permission groups map to roles; decorator-based API permission checks

US-004 — Reset my password via a secure email link

Story ID	US-004
As a...	Staff Member
I want to...	reset my password via a secure email link
So that...	I can regain access without involving IT support
Priority / Points	**Must Have** | Story Points: 2
Acceptance Criteria	Reset link expires in 60 minutes; one-time use only; new password must differ from last three passwords; success redirects to login page

US-005 — Remain logged in across browser sessions using a refresh token

Story ID	US-005
As a...	Staff Member
I want to...	remain logged in across browser sessions using a refresh token
So that...	I do not have to log in every time I open the portal
Priority / Points	**Must Have** | Story Points: 2
Acceptance Criteria	Refresh token valid for 30 days; new access token issued silently; refresh token invalidated on logout; all tokens invalidated on password reset

US-006 — Deactivate a staff account immediately

Story ID	US-006
As a...	School Director
I want to...	deactivate a staff account immediately
So that...	a departing employee loses access to school data instantly
Priority / Points	**Should Have** | Story Points: 2
Acceptance Criteria	Deactivated account returns 401 on next request; active sessions invalidated within 60 seconds; audit log records who deactivated and when

US-007 — View a full audit log of all login events and role changes

Story ID	US-007
As a...	System Admin
I want to...	view a full audit log of all login events and role changes
So that...	I can investigate suspicious access and demonstrate compliance
Priority / Points	**Should Have** | Story Points: 3

Acceptance Criteria	Audit log shows: actor, action, timestamp, IP address; filterable by user, action type, and date range; export to CSV available

EPIC-02 School & Student Administration

US-008 — Bulk-import students from a CSV file with name, class, guardian name, and WhatsApp number

Story ID	US-008
As a...	Bursar
I want to...	bulk-import students from a CSV file with name, class, guardian name, and WhatsApp number
So that...	I can onboard an entire school in minutes rather than entering records one by one
Priority / Points	**Must Have** | Story Points: 5
Acceptance Criteria	CSV template downloadable; validation rejects rows with missing name or guardian phone; duplicate student detection by name + class; import summary shows success count and error rows; partial import allowed
Dev Notes	Use pandas for CSV parsing server-side; return error report as downloadable CSV

US-009 — Add a single student with name, date of birth, class, and at least one guardian

Story ID	US-009
As a...	Bursar
I want to...	add a single student with name, date of birth, class, and at least one guardian
So that...	new enrolments are captured immediately without waiting for bulk import
Priority / Points	**Must Have** | Story Points: 3
Acceptance Criteria	Guardian WhatsApp number validated as valid Ugandan format (+256 or 07xx); student assigned a unique ID; student appears in class list immediately; photo upload optional

US-010 — Edit a student’s details including class, guardian contact, and status (active/inactive)

Story ID	US-010
As a...	Bursar
I want to...	edit a student’s details including class, guardian contact, and status (active/inactive)
So that...	records stay accurate as students move classes or guardians change numbers
Priority / Points	**Must Have** | Story Points: 2

Acceptance Criteria	All edits logged with before/after values; WhatsApp number change triggers opt-in re-confirmation to new number; inactive students excluded from fee reminders but records preserved

US-011 — Add multiple guardians to a single student and mark one as primary

Story ID	US-011
As a...	Bursar
I want to...	add multiple guardians to a single student and mark one as primary
So that...	both parents can receive communications and the primary contact is always clear
Priority / Points	**Should Have** | Story Points: 3
Acceptance Criteria	Up to 3 guardians per student; primary guardian receives all automated messages; secondary guardians receive only fee receipts; each guardian can be linked to multiple students (siblings)

US-012 — Create and manage class groups (e.g., P.5A, S.3 West) with a form teacher

Story ID	US-012
As a...	School Admin
I want to...	create and manage class groups (e.g., P.5A, S.3 West) with a form teacher
So that...	students are organised correctly for marks entry, attendance, and fee reporting
Priority / Points	**Must Have** | Story Points: 2
Acceptance Criteria	Class has name, level (Primary/Secondary), stream, academic year, and form teacher; students can be moved between classes; historical class assignments preserved for report cards

US-013 — Configure the academic year calendar with term start and end dates and holiday periods

Story ID	US-013
As a...	School Admin
I want to...	configure the academic year calendar with term start and end dates and holiday periods
So that...	all automated reminders and reports trigger on the correct dates for our school
Priority / Points	**Must Have** | Story Points: 3
Acceptance Criteria	Up to 3 terms per year; term dates define reminder windows and report generation triggers; holiday periods suppress all automated messaging; calendar visible to all staff

Dev Notes	Calendar drives all Celery beat scheduling; changing term dates must reschedule all pending tasks

US-014 — Search for any student by name, ID, or guardian phone number

Story ID	US-014
As a...	Bursar
I want to...	search for any student by name, ID, or guardian phone number
So that...	I can find a student's record instantly during a parent call or visit
Priority / Points	**Must Have** | Story Points: 2
Acceptance Criteria	Search returns results within 500ms for up to 5,000 students; partial name match supported; results show student name, class, and current fee balance; deep link to full record

US-015 — Generate a school enrolment report showing total students by class and gender

Story ID	US-015
As a...	School Director
I want to...	generate a school enrolment report showing total students by class and gender
So that...	I have accurate headcount data for planning and reporting to the district education office
Priority / Points	**Should Have** | Story Points: 3
Acceptance Criteria	Report shows: total active students, breakdown by class, breakdown by gender, year-on-year change; exportable as PDF and Excel; date-stamped for official use

EPIC-03 Fee Collection & Revenue Automation

US-016 — Configure the fee structure for each term with amount, category (tuition, lunch, transport), and due date per class

Story ID	US-016
As a...	Bursar
I want to...	configure the fee structure for each term with amount, category (tuition, lunch, transport), and due date per class
So that...	the system knows exactly what each student owes and when it is due
Priority / Points	**Must Have** | Story Points: 5
Acceptance Criteria	Fee categories are customisable; different amounts can be set per class; optional mandatory/voluntary flag per category; fee structure cloneable from previous term; changes after term start require Director approval
Dev Notes	FeeStructure model linked to AcademicTerm and ClassGroup; FeeRecord generated per student on term activation

US-017 — View the real-time fee balance for any individual student including payments made and outstanding amount

Story ID	US-017
As a...	Bursar
I want to...	view the real-time fee balance for any individual student including payments made and outstanding amount
So that...	I can answer a parent's query instantly and accurately without checking a ledger
Priority / Points	**Must Have** | Story Points: 3
Acceptance Criteria	Balance view shows: total expected, total paid, outstanding, payment history with dates and MoMo references; data current within 30 seconds of any payment; printable receipt history available

US-018 — View a class-level fee collection summary showing how many students have paid in full, partially, and not at all

Story ID	US-018
As a...	Bursar
I want to...	view a class-level fee collection summary showing how many students have paid in full, partially, and not at all
So that...	I can identify which classes have the most outstanding fees and prioritise follow-up

Priority / Points	**Must Have**	Story Points: 3
Acceptance Criteria	Summary table per class: student count, fully paid count, partial count, zero paid count, total collected (UGX), total outstanding (UGX); sortable by outstanding amount; updated in real time

US-019 — Manually record a cash or bank transfer payment for a student

Story ID	US-019
As a...	Bursar
I want to...	manually record a cash or bank transfer payment for a student
So that...	payments made outside MoMo are still captured in the system and the student's balance updated
Priority / Points	**Must Have**	Story Points: 3
Acceptance Criteria	Manual payment requires: amount, payment method (cash/bank/cheque), reference, date, and Bursar ID; triggers WhatsApp receipt to parent; appears in reconciliation report as 'manual'; requires Director approval above UGX 500,000

US-020 — Generate a defaulter list of students with outstanding balances after the fee deadline

Story ID	US-020
As a...	Bursar
I want to...	generate a defaulter list of students with outstanding balances after the fee deadline
So that...	I can systematically follow up with parents who have not paid rather than searching record by record
Priority / Points	**Must Have**	Story Points: 3
Acceptance Criteria	Defaulter list filterable by class, amount range, and days overdue; shows guardian WhatsApp number for easy follow-up; bulk WhatsApp reminder triggerable directly from the list; exportable as PDF and Excel

US-021 — Create a fee payment plan for a parent who cannot pay in full at once

Story ID	US-021
As a...	Bursar
I want to...	create a fee payment plan for a parent who cannot pay in full at once
So that...	the school recovers more revenue from struggling families rather than losing it entirely
Priority / Points	**Must Have**	Story Points: 5

Acceptance Criteria	Payment plan specifies: instalments (up to 4), amounts per instalment, due dates; each instalment triggers a WhatsApp reminder on its due date; plan visible on student's balance view; overdue instalment triggers escalation to Bursar
Dev Notes	PaymentPlan model with Instalment child records; Celery beat schedules reminder per instalment due date

US-022 — View a school-wide fee collection dashboard showing total expected, collected, and outstanding for the current term

Story ID	US-022
As a...	School Director
I want to...	view a school-wide fee collection dashboard showing total expected, collected, and outstanding for the current term
So that...	I know our revenue position at any moment and can take action if collection is falling behind
Priority / Points	**Must Have** | Story Points: 3
Acceptance Criteria	Dashboard shows: total expected (UGX), total collected (UGX), collection rate (%), top 5 defaulting classes, daily collection trend chart for the current term; refreshes every 5 minutes

US-023 — Reverse or correct an incorrectly posted payment with a reason and Director approval

Story ID	US-023
As a...	Bursar
I want to...	reverse or correct an incorrectly posted payment with a reason and Director approval
So that...	errors are corrected without leaving the accounts in an inconsistent state
Priority / Points	**Should Have** | Story Points: 5
Acceptance Criteria	Reversal requires Director approval above UGX 100,000; reversal logged with original payment ID, reason, actor, approver, and timestamp; parent notified via WhatsApp of the correction; balance updated immediately on approval

US-024 — Apply a fee waiver or discount to a specific student's account with a reason

Story ID	US-024
As a...	Bursar
I want to...	apply a fee waiver or discount to a specific student's account with a reason
So that...	scholarship students, staff children, or hardship cases are handled accurately

Priority / Points	**Should Have**	Story Points: 3
Acceptance Criteria	Waiver requires: amount or percentage, category, reason, and Director approval; waiver recorded on student account; reduces outstanding balance; shown distinctly in fee history; reportable separately for audit

US-025 — Compare fee collection rates across terms and academic years

Story ID	US-025
As a...	School Director
I want to...	compare fee collection rates across terms and academic years
So that...	I can identify whether collection is improving and hold the bursar accountable to measurable targets
Priority / Points	**Should Have**	Story Points: 3
Acceptance Criteria	Trend view shows collection rate per term for last 3 academic years; line chart with percentage and absolute UGX values; downloadable as PDF for board reporting

EPIC-04 Mobile Money Integration (MTN & Airtel)

US-026 — Receive an Instant Payment Notification (IPN) from MTN MoMo when a parent pays fees

Story ID	US-026
As a...	System
I want to...	receive an Instant Payment Notification (IPN) from MTN MoMo when a parent pays fees
So that...	the system captures the payment automatically without any manual input from the bursar
Priority / Points	**Must Have** | Story Points: 8
Acceptance Criteria	Webhook endpoint validates HMAC-SHA256 signature; rejects unsigned requests with 401; duplicate reference returns 200 without double-posting; payment matched to student by phone number and reference within 10 seconds; unmatched payment placed in reconciliation queue
Dev Notes	Idempotency key = MoMo reference; atomic DB transaction on FeeRecord update

US-027 — Receive an IPN from Airtel Money when a parent pays fees

Story ID	US-027
As a...	System
I want to...	receive an IPN from Airtel Money when a parent pays fees
So that...	parents on Airtel have the same seamless payment experience as MTN users
Priority / Points	**Must Have** | Story Points: 5
Acceptance Criteria	Same acceptance criteria as US-026; Airtel API uses different authentication scheme — OAuth 2.0 token; network field on Payment record set to 'Airtel'; reconciliation report shows split by network
Dev Notes	Separate Airtel OAuth token refresh task in Celery; token cached in Redis with 1-hour TTL

US-028 — Receive a WhatsApp payment receipt within 60 seconds of my Mobile Money payment being confirmed

Story ID	US-028
As a...	Parent
I want to...	receive a WhatsApp payment receipt within 60 seconds of my Mobile Money payment being confirmed

So that...	I have immediate proof of payment and confidence the school has received my money
Priority / Points	**Must Have**	Story Points: 3
Acceptance Criteria	Receipt sent within 60 seconds of IPN receipt; receipt contains: student name, amount, MoMo reference, date, school name, remaining balance; receipt delivery confirmed via WhatsApp read receipt logging; retry once if delivery fails within 5 minutes

US-029 — See a reconciliation queue of unmatched Mobile Money payments that the system could not automatically link to a student

Story ID	US-029
As a...	Bursar
I want to...	see a reconciliation queue of unmatched Mobile Money payments that the system could not automatically link to a student
So that...	no money is lost or left unposted even if a parent used the wrong reference
Priority / Points	**Must Have**	Story Points: 5
Acceptance Criteria	Unmatched payment shows: amount, payer phone number, MoMo reference, timestamp, suggested student matches (by phone number lookup); Bursar can accept a suggestion or manually search and link; every manual link logged in audit trail
Dev Notes	Suggest matches by looking up guardian phone number in Guardians table; fuzzy match if exact fails

US-030 — View and download a daily payment reconciliation report

Story ID	US-030
As a...	Bursar
I want to...	view and download a daily payment reconciliation report
So that...	I can confirm that all MoMo receipts match the system records before closing the day's books
Priority / Points	**Must Have**	Story Points: 3
Acceptance Criteria	Report shows: total payments received (MTN + Airtel), total auto-matched, total manually matched, total unmatched, total UGX collected; generated daily at 11:00 PM EAT; emailed to Bursar and Director; downloadable as PDF and Excel from portal

US-031 — Configure the school's MTN and Airtel merchant credentials and webhook endpoints from the admin panel

Story ID	US-031

As a...	System Admin
I want to...	configure the school's MTN and Airtel merchant credentials and webhook endpoints from the admin panel
So that...	each school's MoMo integration is isolated and there is no cross-school payment risk
Priority / Points	**Must Have** | Story Points: 5
Acceptance Criteria	Credentials stored encrypted at rest (AES-256); webhook URL generated per school with unique signing key; test payment endpoint available for credential validation; credential rotation does not interrupt live payments
Dev Notes	Credentials stored in SchoolMoMoConfig model; never returned in API responses in plain text

EPIC-05 WhatsApp Communication Layer

US-032 — Send an automated fee reminder to a guardian via WhatsApp 14, 7, 3, and 1 day before the fee deadline

Story ID	US-032
As a...	System
I want to...	send an automated fee reminder to a guardian via WhatsApp 14, 7, 3, and 1 day before the fee deadline
So that...	parents are consistently reminded without the bursar having to manually contact anyone
Priority / Points	**Must Have** | Story Points: 5
Acceptance Criteria	Reminder schedule driven by term fee deadline in AcademicTerm; message includes: student name, amount due, deadline, payment instructions; reminders suppressed if balance already zero; Bursar can pause reminders for specific students; delivery status logged
Dev Notes	Celery beat task per term; schedule recalculated if term dates change; WhatsApp message template approved via Meta Business Manager

US-033 — Send BALANCE to the school’s WhatsApp number and receive my child’s current fee balance instantly

Story ID	US-033
As a...	Parent
I want to...	send BALANCE to the school’s WhatsApp number and receive my child’s current fee balance instantly
So that...	I can check what I owe at any time without calling the school or visiting in person
Priority / Points	**Must Have** | Story Points: 5
Acceptance Criteria	Response within 5 seconds of message receipt; response shows: student name, term, total fees, amount paid, outstanding balance, and next payment deadline; if guardian has multiple children, response lists all; BALANCE works 24/7
Dev Notes	Inbound webhook from WhatsApp Cloud API; phone number looked up in Guardians table; if number not found, reply with registration instructions

US-034 — Send RECEIPT to the school’s WhatsApp number and receive my most recent payment receipt

Story ID	US-034
As a...	Parent

I want to...	send RECEIPT to the school's WhatsApp number and receive my most recent payment receipt
So that...	I have a digital record of my payment without needing to visit the school office
Priority / Points	**Must Have** | Story Points: 3
Acceptance Criteria	Response includes: payment date, amount, MoMo reference, school name, and remaining balance; if no payment on record, a helpful message is returned; RECEIPT [date] returns receipt for a specific date if supplied

US-035 — Send a broadcast WhatsApp message to all guardians, a specific class, or a custom group

Story ID	US-035
As a...	Bursar
I want to...	send a broadcast WhatsApp message to all guardians, a specific class, or a custom group
So that...	I can reach parents with school announcements, emergency notices, or event reminders without using personal WhatsApp
Priority / Points	**Must Have** | Story Points: 5
Acceptance Criteria	Broadcast targets: all school, by class, by fee status, or custom list; message preview before send; scheduled send supported (e.g., send at 8:00 AM tomorrow); delivery report shows sent, delivered, read, and failed counts; WhatsApp template compliance enforced
Dev Notes	Bulk sends go through a Celery queue with rate limiting to respect WhatsApp API limits (1,000 conversations/day on basic tier)

US-036 — Send HELP to the school's WhatsApp number and receive a list of available commands

Story ID	US-036
As a...	Parent
I want to...	send HELP to the school's WhatsApp number and receive a list of available commands
So that...	I can discover what I can do without needing to contact the school or read a manual
Priority / Points	**Must Have** | Story Points: 2
Acceptance Criteria	HELP returns a formatted list of all commands with a one-line description of each; message is translated to the school's configured language (English default, Luganda optional); HELP works 24/7

US-037 — Automatically send an attendance alert to a guardian when their child is marked absent

Story ID	US-037
As a...	System
I want to...	automatically send an attendance alert to a guardian when their child is marked absent
So that...	parents are notified immediately and can contact the school if the absence is unexpected
Priority / Points	**Must Have** | Story Points: 3
Acceptance Criteria	Alert sent within 30 minutes of roll-call close; message includes: student name, date, class; suppressed if absence was pre-notified by parent; if WhatsApp delivery fails after 15 minutes, SMS fallback triggered automatically

US-038 — Send PLAN to the school's WhatsApp number to request a payment plan for outstanding fees

Story ID	US-038
As a...	Parent
I want to...	send PLAN to the school's WhatsApp number to request a payment plan for outstanding fees
So that...	I can start the process of arranging a payment plan without visiting the school in person
Priority / Points	**Should Have** | Story Points: 5
Acceptance Criteria	PLAN triggers a notification to the Bursar with guardian name and outstanding balance; Bursar receives an in-portal task to call or WhatsApp the parent and set up a plan; parent receives acknowledgement message within 2 minutes; response time SLA is 1 business day

US-039 — View a WhatsApp message delivery dashboard showing sent, delivered, read, and failed counts per broadcast

Story ID	US-039
As a...	Bursar
I want to...	view a WhatsApp message delivery dashboard showing sent, delivered, read, and failed counts per broadcast
So that...	I can confirm parents are receiving our messages and identify guardians with non-functional numbers
Priority / Points	**Should Have** | Story Points: 3

Acceptance Criteria	Dashboard shows delivery metrics per broadcast with timestamp; failed deliveries show reason code from WhatsApp API; one-click export of failed numbers for manual follow-up; data retained for 90 days

US-040 — Configure automated WhatsApp message templates for different event types with school name and custom text

Story ID	US-040
As a...	School Admin
I want to...	configure automated WhatsApp message templates for different event types with school name and custom text
So that...	all outgoing messages are professional, on-brand, and consistent without requiring staff to write them each time
Priority / Points	**Should Have** | Story Points: 3
Acceptance Criteria	Templates editable per message type (reminder, receipt, alert, announcement); variables available: {student_name}, {amount}, {balance}, {date}, {school_name}; templates must be submitted for Meta approval before first use; preview available before activation

EPIC-06 Academic Module & UNEB Report Cards

US-041 — Enter marks for each student in my class by subject and assessment type (exam, test, coursework)

Story ID	US-041
As a...	Teacher
I want to...	enter marks for each student in my class by subject and assessment type (exam, test, coursework)
So that...	all marks are captured digitally and available for report generation without manual aggregation
Priority / Points	**Must Have** | Story Points: 5
Acceptance Criteria	Marks entry by class, subject, and assessment type; validation: marks cannot exceed maximum set for assessment; save progress without submitting; submit locks entry for that assessment (Director can unlock); bulk entry via copy-paste from spreadsheet supported
Dev Notes	AssessmentMark model; marks locked on submission; unlock requires audit log entry

US-042 — Automatically compute each student's total marks, class average, and rank when a teacher submits marks

Story ID	US-042
As a...	System
I want to...	automatically compute each student's total marks, class average, and rank when a teacher submits marks
So that...	all calculations are accurate and instant, eliminating manual aggregation errors
Priority / Points	**Must Have** | Story Points: 3
Acceptance Criteria	Totals computed per subject and overall; class ranking updated when all subjects submitted; tied students receive the same rank; computations run within 10 seconds of submission for classes up to 80 students

US-043 — Generate a UNEB-format Primary 7 (PLE) report card for each student in P7

Story ID	US-043
As a...	Bursar / Admin
I want to...	generate a UNEB-format Primary 7 (PLE) report card for each student in P7
So that...	the school produces legally compliant PLE reports without any manual formatting
Priority / Points	**Must Have** | Story Points: 8

Acceptance Criteria	PLE report card matches UNEB published P7 template exactly: subject names, aggregate scoring, division computation (I–IV, U); includes student photo if uploaded; school stamp/name in header; class teacher and head teacher signature lines; PDF output; batch generation for entire class in one action
Dev Notes	Use ReportLab or WeasyPrint for PDF; UNEB P7 template verified against 2024 UNEB guidelines

US-044 — Generate UNEB-format Senior 4 (UCE) report cards for each S4 student

Story ID	US-044
As a...	Admin
I want to...	generate UNEB-format Senior 4 (UCE) report cards for each S4 student
So that...	the school produces compliant UCE reports with correct division and aggregate computation
Priority / Points	**Must Have** | Story Points: 8
Acceptance Criteria	UCE report matches UNEB S4 template; aggregate computed as best 8 subjects; divisions assigned per UNEB table (I: 4–9, II: 10’19, III: 20–29, IV: 30–36, U: 37+); subject grades (D1–F9) computed correctly; credit counts shown; batch PDF generation for class
Dev Notes	Division table encoded as lookup; verify against current UNEB UCE grading published guidelines

US-045 — Generate UNEB-format Senior 6 (UACE) report cards for each S6 student

Story ID	US-045
As a...	Admin
I want to...	generate UNEB-format Senior 6 (UACE) report cards for each S6 student
So that...	the school produces compliant UACE reports with principal and subsidiary pass computation
Priority / Points	**Should Have** | Story Points: 8
Acceptance Criteria	UACE report matches UNEB S6 template; principal passes (A–E) and subsidiary passes computed; points for UACE aggregate calculated per UNEB table; combination subjects shown correctly; batch PDF generation

US-046 — View a performance summary for my class showing the highest, lowest, and average marks per subject

Story ID	US-046
As a...	Teacher

I want to...	view a performance summary for my class showing the highest, lowest, and average marks per subject
So that...	I can identify struggling students and subjects before the end of term while there is still time to intervene
Priority / Points	**Should Have** | Story Points: 3
Acceptance Criteria	Summary available per class per term; shows: subject average, highest mark, lowest mark, pass rate (above 50%), and list of students below 40%; visible only to the class teacher and Director; does not require marks to be fully submitted

US-047 — Receive a WhatsApp notification when my child's report card is ready and access it from a link

Story ID	US-047
As a...	Parent
I want to...	receive a WhatsApp notification when my child's report card is ready and access it from a link
So that...	I receive results promptly without having to visit the school or wait for the student to bring a paper copy
Priority / Points	**Should Have** | Story Points: 3
Acceptance Criteria	Notification sent when Director publishes reports for a class; message includes student name, term, and a secure link to PDF (link expires in 30 days); PDF download tracked (viewed/not viewed); manual re-send option for Bursar
Dev Notes	Secure signed URL to S3/Spaces; token-based access without requiring parent login

US-048 — View a school-wide academic performance dashboard for the current term

Story ID	US-048
As a...	School Director
I want to...	view a school-wide academic performance dashboard for the current term
So that...	I can monitor academic standards across all classes and identify underperforming teachers or subjects
Priority / Points	**Nice to Have** | Story Points: 5
Acceptance Criteria	Dashboard shows: average score per class, average per subject school-wide, top 10 and bottom 10 students (anonymised for class teachers), comparison to previous term; accessible to Director only

EPIC-07 Attendance Management

US-049 — Take daily attendance for my class by marking each student present, absent, or late

Story ID	US-049
As a...	Teacher
I want to...	take daily attendance for my class by marking each student present, absent, or late
So that...	attendance is recorded digitally once per day and immediately available to the director without paper registers
Priority / Points	**Must Have** | Story Points: 5
Acceptance Criteria	Attendance entry available from 7:00 AM to 12:00 PM EAT on school days; late submission requires reason; absent students trigger parent alert (see US-037); register locks at 12:00 PM; teacher can view but not edit past registers
Dev Notes	AttendanceRecord model; school-day check against AcademicCalendar before allowing entry

US-050 — View an attendance summary for any class or student for the current term

Story ID	US-050
As a...	School Director
I want to...	view an attendance summary for any class or student for the current term
So that...	I can identify chronic absenteeism patterns and follow up with parents before they become serious
Priority / Points	**Must Have** | Story Points: 3
Acceptance Criteria	Summary shows: attendance rate per student (%), days present, days absent, days late; filterable by class and date range; students below 85% attendance rate flagged in red; exportable as Excel

US-051 — Send an automated alert to the Director when a student's attendance rate falls below 75% in the current term

Story ID	US-051
As a...	System
I want to...	send an automated alert to the Director when a student's attendance rate falls below 75% in the current term
So that...	the Director is proactively notified and can intervene before the student's academic performance is seriously affected
Priority / Points	**Must Have** | Story Points: 3

Acceptance Criteria	Alert triggered when student crosses the 75% threshold; alert shows: student name, class, current attendance rate, number of absences; sent to Director via portal notification and email; one alert per student per week maximum

US-052 — Send a pre-notification via WhatsApp if my child will be absent the next day

Story ID	US-052
As a...	Parent
I want to...	send a pre-notification via WhatsApp if my child will be absent the next day
So that...	the school is informed in advance and my child is not marked as unexpectedly absent
Priority / Points	**Should Have** | Story Points: 3
Acceptance Criteria	Parent sends ABSENT [date] [reason] to school WhatsApp number; teacher receives in-portal notification on their class register for that date; pre-notified absence still recorded but marked as ‘excused’; does not trigger parent alert

US-053 — Cross-reference attendance with fee payment status to identify students who are attending but whose fees are significantly overdue

Story ID	US-053
As a...	Bursar
I want to...	cross-reference attendance with fee payment status to identify students who are attending but whose fees are significantly overdue
So that...	the school can follow up with fee-owing families who are still sending their children, indicating they may have capacity to pay
Priority / Points	**Should Have** | Story Points: 3
Acceptance Criteria	Report shows students with attendance above 80% and fee payment below 50%; sortable by outstanding balance; guardian WhatsApp number shown for easy follow-up; available from the end of Week 4 of each term

EPIC-08 Director & Analytics Dashboard

US-054 — See a single-screen operations overview when I log in showing today's fee collections, attendance rate, and any critical alerts

Story ID	US-054
As a...	School Director
I want to...	see a single-screen operations overview when I log in showing today's fee collections, attendance rate, and any critical alerts
So that...	I start every day with an accurate picture of school operations in under 10 seconds
Priority / Points	**Must Have** | Story Points: 5
Acceptance Criteria	Dashboard shows: today's MoMo collections (UGX), current term collection rate (%), current day's attendance rate (%), count of pending reconciliation items, count of unread alerts; data refreshes every 5 minutes; mobile-responsive layout

US-055 — View a weekly fee collection trend chart for the current term showing daily collections in UGX

Story ID	US-055
As a...	School Director
I want to...	view a weekly fee collection trend chart for the current term showing daily collections in UGX
So that...	I can see whether revenue is flowing steadily or has stalled and take action before the term ends
Priority / Points	**Must Have** | Story Points: 3
Acceptance Criteria	Line chart with one data point per school day; hover tooltip shows date and UGX amount; cumulative vs daily toggle; chart includes target collection line based on expected fees and term length; exportable as PNG for board presentation

US-056 — Generate a term-end financial summary report combining fee collection, MoMo receipts, manual payments, and waivers

Story ID	US-056
As a...	School Director
I want to...	generate a term-end financial summary report combining fee collection, MoMo receipts, manual payments, and waivers
So that...	I have a single auditable document for the school's financial records and board reporting
Priority / Points	**Should Have** | Story Points: 5

Acceptance Criteria	Report includes: total expected fees, total collected (MoMo + manual), total waivers, net collection, collection rate, breakdown by class and fee category; PDF format with school letterhead; date-stamped; Director signature field included

US-057 — Compare my school's fee collection rate to anonymised district averages from the SchoolBridge network

Story ID	US-057
As a...	School Director
I want to...	compare my school's fee collection rate to anonymised district averages from the SchoolBridge network
So that...	I can see how our performance compares to similar schools and set realistic improvement targets
Priority / Points	**Should Have** | Story Points: 5
Acceptance Criteria	Benchmark shows: my school's collection rate vs district median, top quartile, and bottom quartile for the same term; school type and size matched for fairness; school identity of comparators never revealed; available from Week 8 of term onwards

US-058 — Receive a weekly email summary of the school's key metrics every Monday morning

Story ID	US-058
As a...	School Director
I want to...	receive a weekly email summary of the school's key metrics every Monday morning
So that...	I stay informed about school operations even during weeks I cannot access the portal daily
Priority / Points	**Should Have** | Story Points: 2
Acceptance Criteria	Email contains: current term collection rate, last week's MoMo collections, current attendance rate, any critical alerts; sent at 7:00 AM EAT every Monday during active terms; unsubscribable without losing portal access

US-059 — View the number of parent WhatsApp interactions (messages sent, delivered, read) for the current term

Story ID	US-059
As a...	School Director
I want to...	view the number of parent WhatsApp interactions (messages sent, delivered, read) for the current term

So that...	I can assess whether our parent communication strategy is working and identify guardians who are not engaging
Priority / Points	**Nice to Have**	Story Points: 3
Acceptance Criteria	Dashboard panel shows: total messages sent, delivery rate (%), read rate (%), top 10 unresponsive guardians; filterable by message type (reminder, alert, announcement); data refreshes daily

EPIC-09 Automation Engine

US-060 — Execute all scheduled automation tasks (reminders, reports, alerts) reliably even if a worker crashes

Story ID	US-060
As a...	System
I want to...	execute all scheduled automation tasks (reminders, reports, alerts) reliably even if a worker crashes
So that...	no reminders or receipts are lost due to server failures or restarts
Priority / Points	**Must Have** | Story Points: 5
Acceptance Criteria	All tasks use Celery with Redis broker; tasks have retry logic (3 retries with exponential backoff); failed tasks after all retries logged in error queue with payload; dead-letter queue reviewed by System Admin daily; task status visible in admin panel
Dev Notes	Use Celery flower for task monitoring; Redis persistence enabled (AOF) to survive Redis restart

US-061 — Trigger an SMS via Africa's Talking automatically when a WhatsApp message fails delivery after 15 minutes

Story ID	US-061
As a...	System
I want to...	trigger an SMS via Africa's Talking automatically when a WhatsApp message fails delivery after 15 minutes
So that...	critical communications (fee reminders, attendance alerts) reach parents even if they do not have WhatsApp active
Priority / Points	**Must Have** | Story Points: 5
Acceptance Criteria	WhatsApp delivery status checked 15 minutes after send; if not delivered, SMS triggered with same content truncated to 160 characters; SMS delivery status logged; if SMS also fails, event logged for manual follow-up; Bursar sees failed delivery count on dashboard
Dev Notes	WhatsApp webhook delivery status callback; Celery delayed task checks status; Africa's Talking Uganda SMS gateway

US-062 — Generate and email the daily reconciliation report to the Bursar and Director automatically at 11:00 PM EAT

Story ID	US-062
As a...	System

I want to...	generate and email the daily reconciliation report to the Bursar and Director automatically at 11:00 PM EAT
So that...	the school's financial records are reconciled daily without any manual action from staff
Priority / Points	**Must Have** | Story Points: 3
Acceptance Criteria	Report generated as PDF and attached to email; email delivered by 11:15 PM EAT; if generation fails, retry at 11:30 PM and alert System Admin; report also available for manual download in portal at any time

US-063 — Suppress all automated messages during school holidays and weekends as defined in the academic calendar

Story ID	US-063
As a...	System
I want to...	suppress all automated messages during school holidays and weekends as defined in the academic calendar
So that...	parents are not disturbed by school messages during holiday periods, which would damage trust in the platform
Priority / Points	**Must Have** | Story Points: 2
Acceptance Criteria	All outbound message tasks check current date against AcademicCalendar before sending; if date is holiday or weekend, task rescheduled to next school day at 8:00 AM EAT; emergency broadcast (Director-initiated) bypasses suppression
Dev Notes	Single is_school_day() utility function used by all message tasks; calendar cache invalidated when calendar is updated

US-064 — Pause all automated reminders for the entire school for a specified period (e.g., during exams)

Story ID	US-064
As a...	School Director
I want to...	pause all automated reminders for the entire school for a specified period (e.g., during exams)
So that...	I can suppress automated messages at sensitive times without deactivating the system entirely
Priority / Points	**Should Have** | Story Points: 2
Acceptance Criteria	Pause period set with start date, end date, and reason; all reminder tasks check pause flag before executing; Director receives confirmation when pause is active; tasks scheduled during pause period are dropped (not queued); Director receives summary of dropped tasks when pause ends

EPIC-10 Offline & Connectivity Resilience

US-065 — Enter attendance and student marks in the admin portal even when the school's internet connection is down

Story ID	US-065
As a...	Teacher
I want to...	enter attendance and student marks in the admin portal even when the school's internet connection is down
So that...	school operations are not blocked by connectivity failures, which are common in our area
Priority / Points	**Must Have** | Story Points: 8
Acceptance Criteria	Service Worker caches attendance and marks entry pages; data saved to IndexedDB locally; sync indicator shows pending items count; on reconnection, all pending items sync automatically within 60 seconds; conflict resolution: server record wins for already-submitted items, local record wins for drafts
Dev Notes	Workbox for Service Worker; IndexedDB via Dexie.js; sync uses background-sync API with fallback to on-visibility sync

US-066 — View student fee balances from a locally cached version of the portal when the internet is unavailable

Story ID	US-066
As a...	Bursar
I want to...	view student fee balances from a locally cached version of the portal when the internet is unavailable
So that...	I can still answer parent queries during connectivity outages without turning them away
Priority / Points	**Must Have** | Story Points: 5
Acceptance Criteria	Fee balance data cached on last successful sync (up to 24 hours old); cache age shown clearly with timestamp; cached data is read-only; write operations (payment entry) queued for sync on reconnection; cache updated automatically on every page load when online

US-067 — Install the SchoolBridge admin portal on my Android phone as a Progressive Web App without visiting an app store

Story ID	US-067
As a...	Teacher
I want to...	install the SchoolBridge admin portal on my Android phone as a Progressive Web App without visiting an app store

So that...	I can take attendance and enter marks from my phone without needing a laptop or app store access
Priority / Points	**Should Have**	Story Points: 5
Acceptance Criteria	PWA manifest configured with school name, icon, and theme colour; install prompt appears after second visit; installed app launches in standalone mode (no browser chrome); push notification permission requested for attendance reminders; works on Android 8+ and Chrome 80+
Dev Notes	manifest.json with all required fields; beforeinstallprompt event handled; icons in 192px and 512px

US-068 — Automatically retry failed MoMo webhook deliveries up to 5 times with exponential backoff before marking as failed

Story ID	US-068
As a...	System
I want to...	automatically retry failed MoMo webhook deliveries up to 5 times with exponential backoff before marking as failed
So that...	transient network failures between MoMo APIs and SchoolBridge do not cause missed payment records
Priority / Points	**Should Have**	Story Points: 3
Acceptance Criteria	Retry intervals: 1 min, 2 min, 5 min, 10 min, 30 min; each retry logged with timestamp and response code; after 5 failures, event placed in dead-letter queue and System Admin alerted; successful retry removes from dead-letter queue

EPIC-11 Security & Compliance

US-069 — Enforce HTTPS on all endpoints and redirect all HTTP requests to HTTPS

Story ID	US-069
As a...	System
I want to...	enforce HTTPS on all endpoints and redirect all HTTP requests to HTTPS
So that...	all data in transit is encrypted and man-in-the-middle attacks are prevented
Priority / Points	**Must Have** | Story Points: 2
Acceptance Criteria	HTTP to HTTPS redirect returns 301; HSTS header with max-age 31536000 set on all responses; SSL certificate auto-renewed via Let's Encrypt; certificate expiry monitored with 30-day advance alert

US-070 — Encrypt all personally identifiable information (PII) fields at rest in the database

Story ID	US-070
As a...	System
I want to...	encrypt all personally identifiable information (PII) fields at rest in the database
So that...	a database breach does not expose parent phone numbers, student names, or payment data in plain text
Priority / Points	**Must Have** | Story Points: 5
Acceptance Criteria	Encrypted fields: guardian phone number, student date of birth, guardian name, payment MoMo reference; encryption uses AES-256 via django-fernet-fields; encryption keys stored in environment variables (not in code or DB); key rotation procedure documented
Dev Notes	django-fernet-fields; key rotation requires re-encryption job — document the procedure in ops runbook

US-071 — Log every payment event, data change, and access event in an immutable audit trail

Story ID	US-071
As a...	System
I want to...	log every payment event, data change, and access event in an immutable audit trail
So that...	the school can demonstrate compliance, investigate disputes, and detect unauthorised access
Priority / Points	**Must Have** | Story Points: 3

Acceptance Criteria	Audit log records: actor (user ID + role), action type, affected record ID, before/after values (for edits), IP address, timestamp; audit log is append-only (no update or delete permitted); retention: 3 years; searchable by actor, record type, and date range

US-072 — Rate-limit all API endpoints to prevent brute-force and denial-of-service attacks

Story ID	US-072
As a...	System
I want to...	rate-limit all API endpoints to prevent brute-force and denial-of-service attacks
So that...	the platform remains available to legitimate users even under automated attack
Priority / Points	**Must Have** | Story Points: 3
Acceptance Criteria	Authentication endpoints: 10 requests per minute per IP; general API: 100 requests per minute per authenticated user; MoMo webhook: 500 per minute per school (higher for payment events); rate limit headers returned (X-RateLimit-Remaining); blocked IPs logged and alerted if block count exceeds 50 per hour

US-073 — Receive an automated alert when any of the following occur: failed webhook delivery exceeding 10 in one hour, database backup failure, certificate expiry within 30 days, or Celery queue depth exceeding 1,000 tasks

Story ID	US-073
As a...	System Admin
I want to...	receive an automated alert when any of the following occur: failed webhook delivery exceeding 10 in one hour, database backup failure, certificate expiry within 30 days, or Celery queue depth exceeding 1,000 tasks
So that...	infrastructure issues are detected and resolved before they affect schools and parents
Priority / Points	**Must Have** | Story Points: 3
Acceptance Criteria	Alerts sent via email and SMS; alert includes: type, timestamp, affected component, and recommended action; alert acknowledged in admin panel; repeated alerts suppressed for 1 hour after acknowledgement; all alert history retained for 90 days

EPIC-12 In-App Notifications & Staff Alerts

US-074 — Receive an in-app notification when a new Mobile Money payment is received

Story ID	US-074
As a...	Bursar
I want to...	receive an in-app notification when a new Mobile Money payment is received
So that...	I am immediately aware of incoming payments during school hours without refreshing the page
Priority / Points	**Must Have** | Story Points: 3
Acceptance Criteria	Notification appears in portal notification bell within 30 seconds of IPN receipt; notification shows: student name, amount, network (MTN/Airtel); clicking opens the student's fee record; notification marked read on click; unread count shown on bell icon
Dev Notes	Use Django Channels (WebSocket) or Server-Sent Events for real-time push; fallback to 30-second polling if WebSocket unavailable

US-075 — Receive an in-app alert when a payment in the reconciliation queue has been waiting unmatched for more than 24 hours

Story ID	US-075
As a...	Bursar
I want to...	receive an in-app alert when a payment in the reconciliation queue has been waiting unmatched for more than 24 hours
So that...	no money sits unallocated for more than one business day, reducing the risk of it being overlooked permanently
Priority / Points	**Must Have** | Story Points: 2
Acceptance Criteria	Alert triggers at 24-hour mark from IPN receipt; shows: amount, payer phone, MoMo reference, time waiting; alert escalates to Director if unresolved after 48 hours; resolved alerts removed from active alert list

US-076 — Receive an in-app reminder to submit attendance if I have not done so by 10:00 AM on a school day

Story ID	US-076
As a...	Teacher
I want to...	receive an in-app reminder to submit attendance if I have not done so by 10:00 AM on a school day
So that...	attendance records are complete every day without the Director having to chase individual teachers

Priority / Points	**Should Have**	Story Points: 2
Acceptance Criteria	Reminder appears at 10:00 AM EAT if no attendance submitted for that teacher's class that day; reminder shown as a banner on any portal page; link goes directly to attendance entry for today; suppressed on non-school days; logged for Director visibility

US-077 — Receive a daily morning digest at 7:30 AM showing yesterday's collections, today's pending tasks, and any unresolved alerts

Story ID	US-077
As a...	Director
I want to...	receive a daily morning digest at 7:30 AM showing yesterday's collections, today's pending tasks, and any unresolved alerts
So that...	I start each working day with a complete situational awareness without logging into the portal first
Priority / Points	**Should Have**	Story Points: 3
Acceptance Criteria	Digest delivered via email and WhatsApp; email includes: yesterday's MoMo collections (UGX), unresolved reconciliation items, teachers with missing attendance, fee reminders sent, any system alerts; WhatsApp version is a short summary (under 300 characters) with link to full portal; send time configurable per school

SchoolBridge — Development Backlog | End of Document

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